



UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES
DEHRADUN

Applied DevOps

Experiment 2

MTECH - School of Computer Science
CYBER SECURITY AND FORENSICS

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Objective — Exploring Application Deployment and Release Automation (ARA) in DevOps using Free ARA Tools

Problem Statement:

To demonstrate how Application Deployment and Release Automation (ARA) supports Continuous Delivery using free and open-source tools such as Git, GitHub Actions, and Docker.

Tools and Environment :

Development Environment

- Operating System: macOS (MacBook Air)
IDE: Visual Studio Code
- Terminal: zsh shell
- Web Browser: Google Chrome

Version Control

- Git: Used for tracking source code changes and managing versions
- GitHub: Cloud-based repository hosting platform
Authentication Method: SSH key-based authentication

CI/CD Automation

- GitHub Actions: Used to automate the build process whenever code is pushed to the main branch
- Workflow File Location: **.github/workflows/ci.yml**

Containerization & Deployment

- Docker Desktop for Mac: Installed and running locally
- Docker Engine: Used to build container images
- Base Image Used: **nginx:alpine**
- Deployment Method: Docker container deployed locally using port mapping
- Application Access URL:
http://localhost:8080

Task to be Performed

Using **free and open-source ARA tools**, design and perform a simple experiment to demonstrate how deployment automation supports **CD**.

Perform the following:

1. Create a simple application (e.g., a basic HTML page or “Hello World” web app).
2. Use **Git** (free) as a version control system to store the application source code.
3. Configure **Jenkins (Community Edition)** or **GitHub Actions** to automatically build the application whenever a code change is pushed to the repository.
4. Use **Docker** to package the application into a container image.
5. Automate the deployment of the containerized application to a local environment using **Docker** or **Docker Compose**.
6. Trigger the entire pipeline automatically on every code change.
7. Observe how automation reduces manual effort and speeds up the release process.

Step-by-Step Procedure:

1. Install necessary dependencies

```
sheoraninfosec — git-remote-https — bash -c #!/bin/bash\012\012# We don't need return codes for "$({command})", only...

Last login: Thu Jan 22 12:40:17 on ttys004
sheoraninfosec@Jigeshs-MacBook-Air ~ % /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
==> Checking for `sudo` access (which may request your password)...
Password:
==> This script will install:
/opt/homebrew/bin/brew
/opt/homebrew/share/doc/homebrew
/opt/homebrew/share/man/man1/brew.1
/opt/homebrew/share/zsh/site-functions/_brew
/opt/homebrew/etc/bash_completion.d/brew
/opt/homebrew
/etc/paths.d/homebrew

Press RETURN/ENTER to continue or any other key to abort:
==> /usr/bin/sudo /usr/sbin/chown -R sheoraninfosec:admin /opt/homebrew
==> Downloading and installing Homebrew...
remote: Enumerating objects: 8976, done.
remote: Counting objects: 100% (3804/3804), done.
remote: Compressing objects: 100% (45/45), done.
remote: Total 8976 (delta 3774), reused 3761 (delta 3759), pack-reused 5172 (from 2)

[sheoraninfosec@Jigeshs-MacBook-Air ~ % brew install git
git 2.51.0 is already installed but outdated (so it will be upgraded).
==> Fetching downloads for: git
✓ Bottle Manifest git (2.53.0)
✓ Bottle Manifest libiconv (1.18)
```

2. Created a basic HTML file named **index.html**.

<> index.html X Dockerfile

<> index.html > ...

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 |   <title>DevOps ARA Experiment</title>
5 </head>
6 <body>
7 |   <h1>Hello World - Automated Deployment!</h1>
8 |   <p>This app is deployed automatically using GitHub Actions and Docker.</p>
9 </body>
10 </html>
11
```

3. Created a Dockerfile to containerize the application.

<> index.html

Dockerfile X

Dockerfile > ...

```
1 FROM nginx:alpine
2 COPY index.html /usr/share/nginx/html/index.html
3 EXPOSE 80
4
```

4. Build docker image

● sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % docker build -t myapp .

```
[+] Building 12.4s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 114B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 269B
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a
=> => resolve docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a
=> => sha256:88799a707571093c0492a622acc0dbaae6337f23bd486eadf27ca684def26a7a 1.40kB / 1.40kB
=> => sha256:da8475fa07c770fbfd5a1bbb5cbd8c44fea3c125feb65eb76a1c199c85184bd 19.71MB / 19.71MB
=> => sha256:7833e4e4252caed7bfc2fa1e0dfb929d975482f39ccc04e1444ab7f121cc882d 1.21kB / 1.21kB
=> => sha256:821790ca706fdc0c24606a10201118cec0f6b3051da3b31366cf0e62ec23ab67 402B / 402B
=> => sha256:9084d2ffc28327306889767de8533d0d890e8e540797ce9933a917927dde87fd 953B / 953B
=> => sha256:31d394b0c9eda37c098db2471823834f196fc546d924d31f47e554b378133051 622B / 622B
=> => sha256:c5ad07fbd6e655e50b0a471f744c9373664dff6236d23b1c2d4c37903012296 1.87MB / 1.87MB
=> => sha256:d8ad8cd72600f46cc068e16c39046ebc76526e41051f43a8c249884b200936c0 4.20MB / 4.20MB
=> => extracting sha256:d8ad8cd72600f46cc068e16c39046ebc76526e41051f43a8c249884b200936c0
=> => extracting sha256:c5ad07fbd6e655e50b0a471f744c9373664dff6236d23b1c2d4c37903012296
=> => extracting sha256:31d394b0c9eda37c098db2471823834f196fc546d924d31f47e554b378133051
=> => extracting sha256:9084d2ffc28327306889767de8533d0d890e8e540797ce9933a917927dde87fd
=> => extracting sha256:821790ca706fdc0c24606a10201118cec0f6b3051da3b31366cf0e62ec23ab67
=> => extracting sha256:7833e4e4252caed7bfc2fa1e0dfb929d975482f39ccc04e1444ab7f121cc882d
=> => extracting sha256:88799a707571093c0492a622acc0dbaae6337f23bd486eadf27ca684def26a7a
=> => extracting sha256:da8475fa07c770fbfd5a1bbb5cbd8c44fea3c125feb65eb76a1c199c85184bd
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:78dd9f0be51661149a4d00be1529afd684fdf208fa45440f9f614c9b5aade3f4
=> => exporting config sha256:514bcc31537975eb9815f810ac9dd894ab9e65a9a3c3bf7449f56b915a561db6
=> => exporting attestation manifest sha256:11daaa6a430b5f5914f87a8a1ec9e4b12836a855ed62a4d5681a9a6d03599961
=> => exporting manifest list sha256:5edda6a5a6d142c756107bbb0cc76b57bfe72f22465742d125938f6d42214ddc
=> => naming to docker.io/library/myapp:latest
=> => unpacking to docker.io/library/myapp:latest
```

5. Run the docker container

- sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % docker run -d -p 8080:80 myapp
e8ac94104dc8db41c263939c8fdb94285c5b158529177f00935de35fe51d244a

6. Application accessed at:

Hello World - Automated Deployment!

This app is deployed automatically using GitHub Actions and Docker.

7. Docker ps command provides detailed information about the process that is currently running.

8. Use Docker stop <process id> to stop the process.

```
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
09192ed355dd   myapp     "/docker-entrypoint...." About a minute ago Up About a minute 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp loving_booth
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % docker stop 09192ed355dd
09192ed355dd
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment %
```

9. Make a little change to the code and build and run docker file to verify if everything is running smoothly before pushing it to github.

<> index.html > html > body > h1

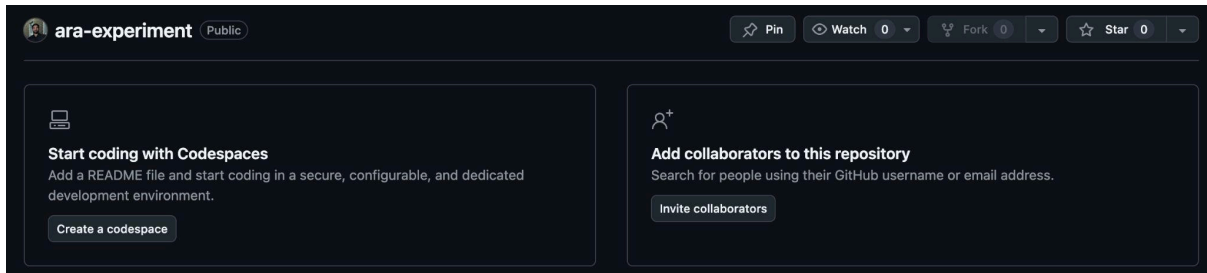
```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>DevOps ARA Experiment</title>
5  </head>
6  <body>
7  |   <h1>Hello World - Version 2</h1>
8  |   <p>This app is deployed automatically using GitHub Actions and Docker.</p>
9  </body>
10 </html>
11
```

```
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % docker build -t myapp .
[+] Building 1.9s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 114B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 257B
=> CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
=> => resolve docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
=> exporting to image
=> exporting layers
=> => exporting manifest sha256:b9dae063d05bda5307800c5b7c431e8c1f872bfb6cee1e7545b57866bb3009de
=> => exporting config sha256:d7b550ed8cf3147fd00c9b594f1a781efc3f6c55ce8c2bb48939628b1d1c12d7
=> => exporting attestation manifest sha256:f643e9fd9b3c0772eb1bd6cbb7ece5851d5985b3b1b3f12c958782dd93ba7660
=> => exporting manifest list sha256:2c2673328ddfec9ddd49001661f3c13c52a25da2a08e6d9937acb0196299c65c
=> => naming to docker.io/library/myapp:latest
=> => unpacking to docker.io/library/myapp:latest
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment %
```


Hello World - Version 2

This app is deployed automatically using GitHub Actions and Docker.

10. Make a new github repository named as ara-experiment.



11. Initialize git repository.

```
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/sheoraninfosec/Documents/Documents - Jigeshs's MacBook Air/Masters UPES/Semester 2/Applied Devops/LA
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git add
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
add
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git add .
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git commit -m "Initial commit - Version 2 working"
[master (root-commit) 5f9e938] Initial commit - Version 2 working
 2 files changed, 13 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 index.html
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git remote add origin https://github.com/sheoraninfosec/ara-experiment.git
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment %
```

12. Push to Github using SSH.

13. Remote changed to SSH instead of HTTPS

```
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git remote -v
origin  https://github.com/sheoraninfosec/ara-experiment.git (fetch)
origin  https://github.com/sheoraninfosec/ara-experiment.git (push)
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git remote set-url origin git@github.com:sheoraninfosec/ara-experiment.git

sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git remote -v
origin  git@github.com:sheoraninfosec/ara-experiment.git (fetch)
origin  git@github.com:sheoraninfosec/ara-experiment.git (push)
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % ssh -T git@github.com

Hi sheoraninfosec! You've successfully authenticated, but GitHub does not provide shell access.
sheoraninfosec@Jigeshs-MacBook-Air ara-experiment %
```

```

● sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git push -u origin main

Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 503 bytes | 503.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sheoraninfosec/ara-experiment.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
○ sheoraninfosec@Jigeshs-MacBook-Air ara-experiment %

```

13. Configure Github Actions Pipeline

14. Now every push to the main branch automatically triggers Docker image build.

```

.github > workflows > ci.yml
1  name: Docker Build Pipeline
2
3  on:
4    push:
5      branches:
6        - main
7
8  jobs:
9    build:
10     runs-on: ubuntu-latest
11
12     steps:
13       - name: Checkout Code
14         uses: actions/checkout@v3
15
16       - name: Build Docker Image
17         run: docker build -t myapp .
18

```

15. Pushing the Pipeline configuration to github.

```

● sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git add .
● sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git commit -m "Added GitHub Actions CI pipeline"
[main 25e130c] Added GitHub Actions CI pipeline
2 files changed, 17 insertions(+)
create mode 100644 .DS_Store
create mode 100644 .github/workflows/ci.yml
● sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 943 bytes | 943.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sheoraninfosec/ara-experiment.git
 5f9e938..25e130c  main -> main
○ sheoraninfosec@Jigeshs-MacBook-Air ara-experiment %

```

16. Now we will make some more changes to the html file to check if the pipeline is running smoothly or not.

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>DevOps ARA Experiment</title>
5  </head>
6  <body>
7  |   <h1>Hello World - Version 3</h1>
8  |   <p>This app is deployed automatically using GitHub Actions and Docker.</p>
9  </body>
10 </html>
11

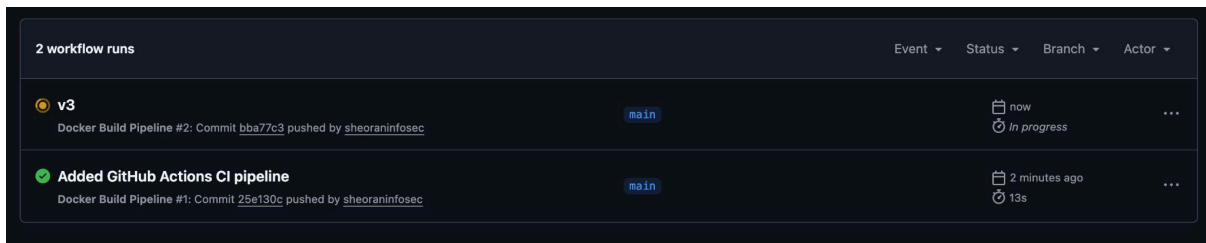
```

17. Committing and pushing the changes to github to trigger continuous deployment.

```

● sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git add .
● sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git commit -m "v3"
[main bba77c3] v3
 2 files changed, 1 insertion(+), 1 deletion(-)
● sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 469 bytes | 469.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:sheoraninfosec/ara-experiment.git
   25e130c..bba77c3  main -> main
○ sheoraninfosec@Jigeshs-MacBook-Air ara-experiment %

```



18. Making changes once again to verify that our pipeline is working smoothly and is not broken.

```

5  </main>
6  <body>
7  |   <h1>Hello World - Version 3.1.0</h1>
8  |   <p>This app is deployed automatically using GitHub Actions and Docker.</p>
9  </body>

```

19. Committing and pushing the changes to github.

- sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git add .
- sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git commit -m "v3.1.0"
[main 4691d8d] v3.1.0
1 file changed, 1 insertion(+), 1 deletion(-)
- sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 284 bytes | 284.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:sheoraninfosec/ara-experiment.git
bba77c3..4691d8d main -> main
- sheoraninfosec@Jigeshs-MacBook-Air ara-experiment % █

20. As we can observe that the github pipeline is working smoothly.

3 workflow runs

Event ▾	Status ▾	Branch ▾	Actor ▾
 v3.1.0		Docker Build Pipeline #3: Commit	main
4691d8d pushed by sheoraninfosec	Today at 12:32 AM ...	23s	
 v3		Docker Build Pipeline #2: Commit	main
bba77c3 pushed by sheoraninfosec	Today at 12:28 AM ...	19s	
 Added GitHub Actions CI ...		Docker Build Pipeline #1: Commit	main
25e130c pushed by sheoraninfosec	Today at 12:26 AM ...	13s	

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